

Khushaan Saini

Software Engineering undergraduate experienced in designing, developing, and integrating scalable, maintainable systems using modern software engineering principles and design patterns. Skilled in RESTful API development, backend architecture, and Agile-based product development, with a focus on scalability, performance, and clean code. Additional applied experience in AI/ML pipelines and LLM orchestration (LangChain, RAG).

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EDUCATION

Thapar Institute of Engineering and Technology

B.E. Electronics and Computer - 8.5/10

July 2023 - July 2027

MDS Senior Secondary School

Higher Secondary (12th CBSE) - 81%

June 2022 - July 2023

PROJECTS

Blog Generation Service – FastAPI Backend [↗](#)

- Built a modular FastAPI backend with dual pipelines (topic-to-content, multilingual routing), generating 600–800-word content in under 3 seconds.
- Used Pydantic validation and end-to-end tracing for 100% API observability, reflecting production-grade practices.
- **Tech:** Python, FastAPI, Pydantic, LangGraph, LangChain, Groq API, LangSmith

Multi-Service Conversational System – Agentic Architecture [↗](#)

- Designed a 3-pipeline multi-service architecture with conditional routing and autonomous API integration (Tavily), cutting processing time by 70%.
- Delivered a Streamlit UI with streamed responses, reducing latency from minutes to under 30 seconds.
- **Tech:** Python, LangGraph, LangChain, Groq API, Tavily, Streamlit

RAG-based Conversational Q&A System [↗](#)

- Built a dual vector store (FAISS + ChromaDB) retrieval pipeline over 500+ chunks with 90%+ precision and sub-800ms responses.
- Added multi-turn memory and a zero-setup Streamlit UI via LangChain LCEL.
- **Tech:** Python, LangChain, LCEL, FAISS, ChromaDB, Groq API, Hugging Face Embeddings, Streamlit

SKILLS

- **Languages & Core CS:** Python, C++, C, SQL, DSA, OOP, Git
- **Backend & Systems:** FastAPI, REST APIs, Streamlit, Jupyter Notebook
- **Data & ML:** Scikit-learn, NumPy, Pandas, Matplotlib
- **Applied AI:** LangChain, LCEL, LangGraph, Hugging Face, RAG, FAISS, ChromaDB, Groq API, Prompt Engineering, LangSmith

CERTIFICATE

Certified AI Foundations Associate – Oracle

Accelerating End-to-End Data Science Workflows – NVIDIA

Data Structures and Algorithms in C++ – Udemy

Generative AI with LangChain and Hugging Face – Udemy

Agentic AI with LangGraph and LangChain – Udemy

AREA OF INTEREST

- Software Engineering & System Design, Backend Development, Generative AI, Agentic AI, Retrieval-Augmented Generation, Deep Learning